

SILMARILS Evaluation for Alberta Buck Identity Binding

Information-Theoretic Tools for Quantum-Resistant Identity Proofs

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1 What SILMARILS Is

SILMARILS is a cryptographic signature scheme whose security is *Information-Theoretic* (IT) – it depends on the mathematics of finite fields, not on computational hardness assumptions (discrete log, factorization, LWE) that quantum algorithms could break. Its core contributions:

- A signature scheme built on *2-out-of-2 Shamir Secret Sharing* over $\mathbb{F}_p$, combined with HMAC and hash functions. All signatures are *algebraic* – they consist of 5 $\mathbb{F}_p$ scalars.
- Two modes: *two-party* (Transferable Designated Verifier, TDV) and *three-party* (broadcast model with signer, holder, verifier). The TDV mode provides signatures only a designated verifier can verify, with the ability to *simulate* accepting transcripts indistinguishable from real ones – making them non-transferable to third parties.
- Security analysis in the Pure IT model, the classical ROM, and the *Quantum Random Oracle Model* (QROM), with a theorem proving that when the four properties (correctness, unforgeability, transferability, secrecy) are all satisfied with error ε , the protocol is an $O(\varepsilon)$ -secure implementation of the ideal three-party digital signature functionality.

Aspect	SILMARILS
Core primitive	2-out-of-2 SSS over $\mathbb{F}_p$
Security model	Information-theoretic (unconditional)
Quantum security	Yes (no computational assumptions)
Signature size	5 scalars $\mathbb{F}_p$ (~ 160 bytes at 256-bit p)
Key size (long-term)	1 scalar K + 2 public weights
Assumptions	TRNG + hash (ROM/QROM) only
Post-quantum compatible	Yes – no EC, no lattice, no code assumptions

2 The Identity Binding System and Its Quantum Vulnerabilities

The Alberta Buck identity layer uses BN254 (alt_bn128) elliptic curve operations extensively:

Primitive	Quantum Risk	Notes
ElGamal encryption	Shor (ECDL)	Encrypts $M = m \cdot G$ under $pk = sk \cdot G$
Chaum-Pedersen proofs	Shor (ECDL)	Used for <code>verifyApprove</code> , <code>verifyDepositorForIssuer</code>
Okamoto sigma protocols	Shor (ECDL)	<code>verifyDepositCoupling</code> , <code>verifyDepositorBinding</code>
Schnorr signatures	Shor (ECDL)	<code>verifyIssuerSchnorr</code>
BN254 pairing checks	Shor (ECDL)	PS credential verification, Groth16
Groth16 SNARK	Shor + KH	<code>mint_batch</code> , <code>spend_a</code> , <code>identity_membership</code>
Poseidon hashing	Grover	Quantum-safe (hash-based, only quadratic speedup)
Identity scalar $m = H(id)$	Grover	Hash function, quantum-safe

2.1 Elliptic curve operations are the primary attack surface.

A large enough quantum computer running Shor’s algorithm would recover:

- Any discrete log sk from a public key $pk = sk \cdot G$.
- Any identity secret m from a registered ElGamal ciphertext E_{addr} .
- The ability to forge Chaum-Pedersen proofs, Okamoto sigmas, and Schnorr signatures, breaking the *binding* between an account and its identity.
- The ability to decrypt any ElGamal ciphertext, revealing previously private identities from past transactions.

Poseidon hashing and the identity scalar derivation are *not* at comparable risk – Grover’s algorithm on hash preimages gives only a quadratic speedup, and for 256-bit outputs the security margin remains adequate.

3 How SILMARILS’s SSS Foundation Could Help

The key structural insight in SILMARILS is that *Shamir Secret Sharing based commitments can be used directly as information-theoretic substitutes for EC-based sigma protocols*. The 2-out-of-2 SSS over $\mathbb{F}_p$ gives perfect secrecy (each share is uniformly random and independent of the secret) and enables linear verification.

A *witness* committed via SSS shares can be verified by reconstructing a linear relation – no elliptic curve operations needed. The verification constraint is just a check that $V = SSS^{-1}(V_0, V_1) = 0$, which is purely algebraic over $\mathbb{F}_p$.

3.1 The analogues in the identity binding system:

EC-based (current)	SSS-based alternative	Gain
E4: $pk_{\text{dep}} = sk_{\text{dep}}G$	$V = \text{SSS}^{-1}(\dots) = 0$	No EC, IT-secure
E2: $C_d = mG + sk_{\text{dep}}R_d$	Linear SSS check over commitments	No EC, IT-secure
E3: $C_e - P_I = mR_e - bH$	Linear SSS check over commitments	No EC, IT-secure
F1 + F2 (B1 depositor binding)	Linear SSS check over commitments	No EC, IT-secure
Chaum-Pedersen DLEQ	SSS-based DLEQ analogue?	Open question

The fundamental pattern: each EC sigma protocol proves knowledge of scalars (m, sk, b) satisfying a *linear relation* that the verifier checks via EC operations. With SSS, we commit to each witness using Shamir shares, then the verifier reconstructs a value that must equal 0 if and only if the relation holds. This is exactly what SILMARILS does for its 5-element signature: the verifier computes $(V_0, V_1) = \text{SSS}(C)$, where C is an algebraic combination of the committed witness, and accepts only if $\text{SSS}^{-1}(V_0, V_1) = 0$.

4 A Concrete Proposal: SSS-Okamoto Sigmas

The deposit coupling proof (E2/E3/E4) and depositor binding proof (E2/F1/F2) can be *directly* translated into SSS-based sigma protocols. The shape is:

1. The prover knows witnesses (m, sk, b) (or (m, sk, r) for B1).
2. The verifier holds public commitments to these values via SSS shares.
3. The prover sends a SSS-committed response that the verifier reconstructs.
4. The verifier checks that the reconstructed value equals 0.

No elliptic curve operations are needed – everything works over $\mathbb{F}_p$. The security is IT: an adversary with unlimited quantum compute cannot forge a proof because the witness is perfectly hidden in each SSS share.

This yields the following quantum-safe equivalents:

EC operation	Quantum-safe replacement	Cost change
ecMul (EC point mult)	Field multiplication	~identical
ecAdd (EC point add)	Field addition	~identical
pairing check	No equivalent needed (SSS-based)	Eliminated
SPoK over ECDL	SSS-based linear check	+2 field mults

The cost model is favorable: field operations on $\mathbb{F}_p$ are already cheap on the EVM (the MODEXP precompile handles modular exponentiation; field mul/add are a few gas each). The current EIP-196 BN254 precompile adds ~6K gas per ecMul. An SSS-based verification would replace those calls with a handful of field arithmetic operations – no precompile dependency.

5 What Would Need to Change

Component	Change required
Identity point M	Would need SSS-based commitment replaces $M = m \cdot C$
E_addr (registered credential)	ElGamal CT over EC -> SSS-shared credential
Deposit coupling (verifyDepositCoupling)	EC sigma -> SSS linear check
Depositor binding (verifyDepositorBinding)	EC sigma -> SSS linear check
Notes spend paths	Membership circuit stays (Poseidon Merkle, already Q-safe)
Groth16 SNARK	Could be replaced by SSS-based accumulator
Identity Merkle tree	Poseidon-based, already Q-safe
PS credential chain	EC pairing -> SSS-based credential

The cleanest migration path:

1. *Keep* the Poseidon Merkle tree (it's already quantum-safe).
2. *Keep* the identity scalar $m = H(\text{id})$
2. *Replace* the EC-based sigma protocols with SSS-based linear checks.
3. *Replace* the ElGamal encryption of M in E_addr with an SSS-shared commitment.
4. *Replace* the Groth16 membership+tie SNARK with a native Poseidon SSS verifier.

6 Relationship to Existing Work

The wallet's *unilateral_a2.py* already contains IdentityTree, identity_leaf, and the deposit coupling proof structure. The signature is a 3-witness Okamoto sigma over EC. Translating this to SSS replaces the commitment points (A2, A3, A4, P_I) from EC points to SSS-shared scalars, and replaces the verification equations (E2, E3, E4) from EC linear algebra to SSS reconstruction checking that a linear combination equals zero.

The wallet’s `b1_binding.py` is structurally identical and benefits from the same translation.

7 Limitations and Caveats

1. *SILMARILS itself is not directly applicable to the identity binding problem.* It is a signature scheme, not a general-purpose commitment or proof system. What transfers is the *technique*: SSS-based linear verification of committed witnesses. The signature’s 5-element structure does not directly map to the 3-witness Okamoto sigma pattern, but the underlying algebraic principle does.
2. *The ElGamal encryption of identity points.* Replacing this with an SSS-based commitment scheme would require rethinking how a registry delivers a certified identity to a client. The current scheme uses the client’s EC public key to encrypt M. An SSS-based scheme would instead share the identity secret into shares distributed during registration, eliminating the EC dependency.
3. *Groth16 SNARK replacement.* The non-native G1 tie (the remaining circuit work described in identity-axis.org) currently needs BN254 emulated EC arithmetic in circom to prove that the G1 point committed in the on-chain sigma is the same point in the Poseidon Merkle tree. An all-field approach eliminates the need for non-native arithmetic entirely – the Poseidon Merkle leaf and the sigma commitment are both $\mathbb{F}_p$ values, so the tie becomes a native circuit constraint.
4. *EVM feasibility.* Field arithmetic in the EVM is practical via the MODEXP precompile (0x05) for modular reduction and native 256-bit operations for addition. The current EIP-196 precompile dependency for BN254 operations would be eliminated. Gas costs are likely comparable or lower for the SSS-based checks.

8 The Hylland Option (Structural Hybrid)

A practical near-term path is a *hybrid* structure using both the existing EC primitives AND an SSS-based fallback, giving dual security until the quantum threat crystallises:

1. The `E_addr` credential carries *both* the ElGamal ciphertext (EC) and an SSS-based identity commitment (IT-Q-safe).

2. The deposit coupling verifies EITHER the EC sigma *or* the SSS sigma.
3. Governance (setMembershipVerifier) upgrades from EC-only to SSS-only when the quantum threat warrants.

This dual-credential approach is similar to how hybrid KEMs work in TLS 1.3 post-quantum migration.

9 Summary

Factor	Assessment
SILMARILS directly usable	No – it is a signature scheme, not a proof system
SSS technique transferable	Yes – core IT-secure commitment + linear verification
Quantum threat addressed	Yes – eliminates all EC operations from identity binding
EVM cost impact	Likely lower (field ops vs EIP-196 precompiles)
Migration difficulty	High – requires rethinking E_addr, PS credentials, SNARK
Cleanest path	SSS-based Okamoto sigmas for deposit coupling + binding
Poseidon Merkle tree	Already quantum-safe, unchanged
Identity concept $m = H(id)$	Already quantum-safe, unchanged

The *single most impactful application* of the SSS technique: *replacing the EIP-196 Okamoto sigmas* (verifyDepositCoupling, verifyDepositorBinding) with purely algebraic $\mathbb{F}_p$ checks/. This addresses the quantum vulnerability of the on-chain identity binding without touching the off-chain Merkle tree, the identity certificate chain, or the wallet’s cryptographic primitives.

9.1 Why not also replace the EC-based receipt verification?

The unilateral receipt (Section sec:receipt of alberta-buck-notes-unilateral) uses a verifiable decrypt proof – a Chaum-Pedersen DLEQ that *eIss* decrypts under M_{rec} to M_I . This is an EC-based proof. Replacing it with an SSS-based receipt would mean proving, in the clear, that an SSS-shared ciphertext opens to a specific value. This is *possible* (the SSS reconstruction already serves as a linear proof), but requires a different receipt format that is no longer compatible with the current bilateral receipt structure. A separate evaluation for receipt migration is warranted.

10 Open Questions

1. SSS-based sigma vs. Okamoto sigma soundness: Does the SSS-based linear check provide the same *special soundness* (extractability of the witness) as the EC Okamoto sigma? Extraction requires rewinding the prover – this works with SSS commitments too, since the commitment is a uniformly random share that perfectly hides the witness (information-theoretic hiding). Extraction from SSS is simpler than from EC sigma protocols because there is no group-theoretic extraction; the witness is recovered directly from the SSS reconstruction of the response. A formal proof would follow the same template as SILMARILS Theorem 1 but applied to the 3-witness linear relation.
2. Can the identity Merkle tree leaf be the SSS share itself? The tree currently stores $\text{identity_leaf}(M) = \text{Poseidon}(M.x, M.y)$. If M is no longer an EC point but an SSS-sharing of the identity, the leaf would commit to one share of the SSS-shared identity. The Poseidon hash would still be quantum-safe. The non-native G1 tie would become a simple equality check between the Poseidon leaf’s preimage and the SSS-shared value – a native constraint in the circuit.